

Sr.No	Module	Weightage %	Marks Weightage%
1	Module-II	20.00	23.68
2	Module-V	20.00	21.05
3	Module-III	20.00	19.74
4	Module-IV	20.00	19.74
5	Module-I	20.00	15.79
Weightage representation okay : Weightage representation exceeded more than 5% :Weightage representation exceeded more than 10%			

CSIT617

[ET]

Enroll. No. - _____

END SEMESTER EXAMINATION : April-May, 2026

Python for Data Science

Time : 3.00 Hrs

Maximum Marks : 60

Note: Attempt questions from all sections as directed.

Section - A : Attempt any Four questions out of Five . Each question carries 06 marks.

- Q1.

(a)

Explain the difference between Mutable and Immutable objects in Python with examples.

(3)

Delete
- (b)

Describe the use of pass and exit() statements in control flow.

(3)

Delete
- Q2.

(a)

What is the purpose of the pickle module in Python file handling?

(3)

Delete
- (b)

Explain the hierarchy of exceptions in Python.

(3)

Delete
- Q3.

Discuss the Data Science Life Cycle and the importance of each stage.

(6)

Delete
- Q4.

(a)

Define Probability Mass Function (PMF) and Probability Density Function (PDF).

(3)

Delete
- (b)

What is the Central Limit Theorem?

(3)

Delete
- Q5.

Explain the techniques for dealing with missing data and removing duplicates during data preparation.

(6)

Delete

Section – B : Attempt any two questions out of three. Each question carries 10 marks.

- Q6.

(a)

Discuss the various loop control statements (for, while) and the use of break and continue.

(5)

Delete
- (b)

How are lists, tuples, and dictionaries passed as arguments to functions?

(5)

Delete
- Q7.

(a)

Explain the process of plotting Sine and Cosine curves using matplotlib.

(5)

Delete
- (b)

Describe how to extract components from an HTML document using BeautifulSoup.

(5)

Delete
- Q8.

(a)

Discuss Data Wrangling and the importance of handling outliers.

(5)

Delete
- (b)

Explain the concepts of Slicing, Dicing, and Filtering data in Python.

(5)

Delete

Section - C : Compulsory question

- Q9.

(a)

Explain the structure of a Python program and the commonly used built-in functions for Strings.

(4)

Delete
- (b)

Discuss the measures of Central Tendency and Dispersion in descriptive statistics.

(4)

Delete
- (c)

Describe the process of Hypothesis Testing through a one-sample test.

(4)

Delete
- (d)

Explain the process of Data Normalization and its significance in data wrangling.

(4)

Delete